12/03/2021

Code-A

Aakash

Medical|IIT-JEE|Foundations

Regd. Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph. 011-47623456

MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** Time : 3.00 Hrs.

Test - I3

Topics Covered : Complete Syllabus of Class-XI & XII (NEET Pattern)

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.

(ii) Mark should be dark and should completely fill the circle.

(iii) Dark only one circle for each entry.

(iv) Dark the circle in the space provided only.

(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.

(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. A particle is moving with velocity of 0.4 m s^{-1} and is subjected to an acceleration of 0.15 m s^{-2} for 2 seconds in the direction at right angles to direction of initial velocity. The magnitude of final velocity is

   (1) 0.9 m s^{-1} (2) 0.7 m s^{-1}

   (3) 0.5 m s^{-1} (4) 0.6 m s^{-1}

2. A ball is thrown upwards from the ground. It returns back to earth in 8 seconds. Its location, 5 second after throw, is (g = 10 m s^{-2})

   (1) 25 m from the ground

   (2) 75 m from the ground

   (3) 45 m from the ground

   (4) 15 m from the ground

3. A vector of length l is turned through an angle of θ about its tail. What is the change in position vector of its head?

   (1) $l\sin\theta$ (2) $2l\sin\left(\frac{\theta}{2}\right)$

   (3) $2l\cos\left(\frac{\theta}{2}\right)$ (4) $l\sin\left(\frac{\theta}{2}\right)$

4. The displacement of a body of mass 2 kg as a function of time is given as $x = (5 + 2t^2)$ m, where t is in second. The increase in kinetic energy, 2 seconds after the motion is

   (1) 8 J (2) 16 J

   (3) 22 J (4) 64 J

5. A body of mass 10 kg is thrown upward with a velocity 98 m s^{-1}. Simultaneously another body is dropped from a tower. After 2 seconds their momenta are found to be same in magnitude. The mass of freely falling body is (g = 9.8 m s^{-2})

   (1) 4 kg (2) 20 kg

   (3) 40 kg (4) 50 kg

6. The momenta of a body in two perpendicular directions at any time t are given as $P_x = 2t^2 + 6$ and $P_y = \frac{3t^2}{2} + 3$. The magnitude of force acting on the body at t = 2 sec is

   (1) 10 units (2) 20 units

   (3) 25 units (4) 14 units

7. A person weighing 60 kg is in a small boat of mass 140 kg, which is at rest, throws a 5 kg stone in horizontal direction with velocity of 14 m s^{-1}. The velocity of the boat immediately after the throw is

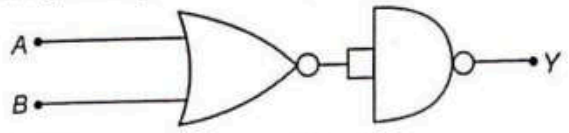
(1) 1.2 m s^{-1}

(2) 0.5 m s^{-1}

(3) 0.35 m s^{-1}

(4) 0.15 m s^{-1}

8. The given logic circuit is equivalent to



(1) NOR (2) AND

(3) OR (4) NAND

9. Which among the following statements, is correct?

(1) Solar cell is used in forward bias for its proper operation

(2) Zener diode, when operated in forward bias, works as voltage regulator

(3) I-V characteristics of solar cell lies in fourth quadrant of coordinate axes

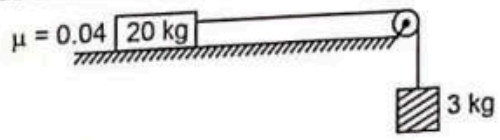
(4) For red LED, material of GaAs is used for diode fabrication.

10. A batsman hits back a ball straight in the direction of bowler without changing its initial speed of 12 m s^{-1}. If the mass of the ball is 300 g, the impulse imparted to the ball (Assume linear motion of the ball).

(1) 3.6 N-s (2) 7.2 N-s

(3) 12.8 N-s (4) 12 N-s

11. What is the acceleration of block placed on rough surface. The mass of block is 20 kg and is connected by light string to hanging mass of 3 kg as shown in figure. (g = 10 m s^{-2})



(1) $\frac{22}{23}$ m s^{-2} (2) $\frac{21}{22}$ m s^{-2}

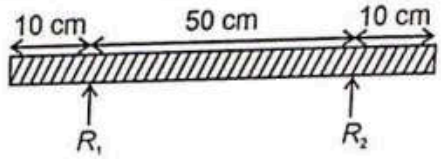
(3) $\frac{16}{19}$ m s^{-2} (4) $\frac{15}{19}$ m s^{-2}

12. A circular race track of radius 300 m is banked at an angle of 15°. If the coefficient of friction between wheels and road is 0.25, the optimum speed of the race car is (tan 15° = 0.26)

(1) 24.3 m s^{-1} (2) 27.9 m s^{-1}

(3) 32.6 m s^{-1} (4) 38.1 m s^{-1}

13. A uniform metal bar 70 cm long and 4 kg in mass is supported on two knife edges placed 10 cm from each end as shown. A 6 kg load is suspended at 30 cm from left end. What is reaction of left support? (g = 9.8 m s^{-2})



(1) 54.88 N (2) 43.19 N

(3) 63.41 N (4) 49.31 N

14. The angular speed of a motor wheel is increased from 20 rps to 52 rps in 16 second. How many revolutions does engine make during this time? (Assuming uniform angular acceleration).

(1) 432 (2) 618

(3) 912 (4) 576

15. A solid cylinder rolls up an inclined plane of inclination 30°. At the bottom of the inclined plane the centre of mass of the cylinder has a speed of 5 m s^{-1}. How long will it take to return to the bottom?

(1) 3 s (2) 4 s

(3) 6 s (4) 2 s

16. The position of a particle at time t is given by an equation $x = \frac{V_0}{A}(1 - e^{At})$, V_0 is constant and $A > 0$. The dimensions of V_0 and A respectively are

(1) M^0LT^{-1}, M^0LT^{-2} (2) M^0LT^{-1}, $M^0L^0T^{-1}$

(3) $M^0L^0T^0$, M^0LT^{-1} (4) M^0LT^{-1}, $M^0L^0T^{-2}$

17. In case of forced oscillation, the resonance peak is large, when the

(1) Damping force is large

(2) Damping force is small

(3) Applied periodic force is small

(4) Restoring force is very small

18. We consider the radiations emitted by a human body. Which of the following statements is true?
(1) The radiation is emitted only during the day
(2) The radiations emitted lies in UV region
(3) The radiations are emitted during summer and absorbed during winters
(4) The radiation emitted is in infrared region

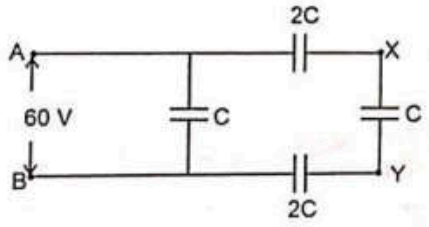
19. For a gas, the difference between two principal specific heats is 4150 J kg^{-1} K^{-1}. What is specific heat of the gas at constant volume if the ratio of specific heat is 1.4?
(1) 8475 J kg^{-1} K^{-1} (2) 10375 J kg^{-1} K^{-1}
(3) 9775 J kg^{-1} K^{-1} (4) 12675 J kg^{-1} K^{-1}

20. Work done in shifting a body of mass M from the centre of the earth to height $h = R$ above the surface of earth is (M = mass of earth, R = radius of earth.)
(1) $\frac{GM^2}{R}$ (2) $\frac{2GM^2}{R}$
(3) $\frac{3}{2}\frac{GM^2}{R}$ (4) $\frac{GM^2}{2R}$

21. Two equal point charges each of 3 μC are separated by a certain distance in meters. If they are located at $(\hat{i}+\hat{j}+\hat{k})$ and $(2\hat{i}+3\hat{j}+3\hat{k})$, then electrostatic force between them is
(1) 9×10^3 N (2) 9×10^{-3} N
(3) 9×10^{-2} N (4) 3×10^{-3} N

22. 5.6 litre of helium gas at STP is adiabatically compressed to 0.7 litre. Taking initial temperature to be T, work done by the gas in the process is
(1) $-RT$ (2) $\frac{-9}{2}RT$
(3) $\frac{-9}{8}RT$ (4) $\frac{-3}{2}RT$

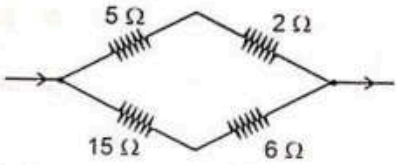
23. A capacitor arrangement is shown below. The potential difference between points X and Y will be



(1) 60 V (2) 30 V
(3) 45 V (4) 12 V

24. Magnetic field due to a current carrying circular loop of radius 12 cm at its centre is 0.5×10^{-3} T. The magnetic field due to this loop at a point on the axis at a distance of 5 cm from centre is
(1) 0.6×10^{-3} T (2) 0.3×10^{-3} T
(3) 0.26×10^{-3} T (4) 0.39×10^{-3} T

25. The resistances in four arms of a Wheatstone bridge network in cyclic order is 5 Ω, 2 Ω, 6 Ω and 15 Ω. If the current of 2.8 A enters the junction of 5 Ω and 15 Ω, then current through 2 Ω resistor is



(1) 2.1 A (2) 3.2 A
(3) 1.6 A (4) 1.9 A

26. What is self-inductance of a coil which induces 5 V, when current in it changes from 3 A to 2 A in a time interval of one millisecond?
(1) 3 mH (2) 5 mH
(3) 6 mH (4) 15 mH

27. A voltage of peak value of 219 V and varying frequency is applied to series LCR combination in which $R = 6$ Ω, $L = 25$ mH and $C = 400$ μF. The frequency of the source at which maximum power is dissipated in circuit is
(1) 30 Hz (2) 25 Hz
(3) 50.3 Hz (4) 60.2 Hz

28. In a sinusoidal wave, time required for a particular point to move from maximum displacement to zero displacement is 0.17 s. The frequency of the wave is
(1) 1.47 Hz (2) 2.62 Hz
(3) 9.52 Hz (4) 11.3 Hz

29. If total time of 10 oscillations of a pendulum is measured as 25 second using a stop watch with least count of 0.5 s, then permissible error in measurement is
(1) 10% (2) 20%
(3) 5% (4) 2%

7. A person weighing 60 kg is in a small boat of mass 140 kg, which is at rest, throws a 5 kg stone in horizontal direction with velocity of 14 m s^{-1}. The velocity of the boat immediately after the throw is

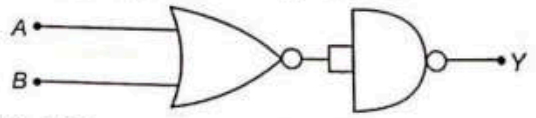
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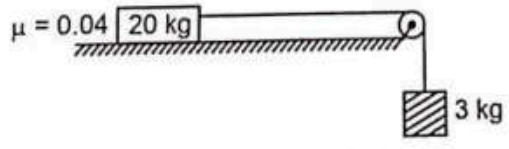
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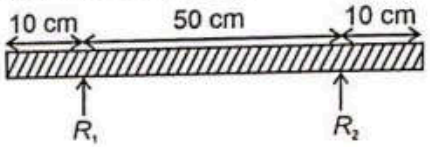
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17. In case of forced oscillation, the resonance peak is large, when the

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(2) Damping force is small

(3) Applied periodic force is small

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30. The threshold frequency of a certain photosensitive metal is ν_0. When it is illuminated by light of frequency $\nu = 2\nu_0$, the maximum velocity of photoelectron is v. What will be maximum velocity of photoelectrons when same metal is illuminated by light of frequency $\nu = 5\nu_0$?

(1) $\sqrt{2}\,v$ (2) $2\,v$

(3) $5\,v$ (4) $3\,v$

31. Water in river 20 m deep is flowing at a speed of 10 m s^{-1}. The shear stress between horizontal layers of water in river in (N m^{-2}) is (η = coefficient of viscosity of water = 10^{-3} SI unit)

(1) 10^{-2} (2) 5×10^{-3}

(3) 10^{-3} (4) 5×10^{-4}

32. Two electric bulbs rated 25 W, 220 V and 100 W, 220 V are connected in series to 300 V supply. Which of the following statement is correct?

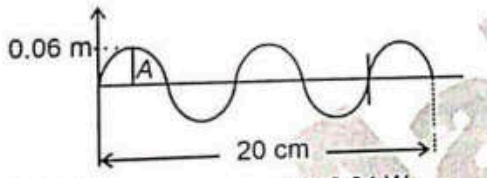
(1) 100 W bulb will fuse

(2) 25 W bulb will fuse

(3) Both 25 W and 100 W will fuse

(4) Neither 25 W nor 100 W will fuse

33. A sinusoidal wave is travelling on a string. If the frequency of the wave is 150 Hz and mass per unit length of the string is 0.2 g m^{-1}. The average power transmitted by this wave shown is



(1) 2.34 W (2) 3.84 W

(3) 4.80 W (4) 6.25 W

34. A horizontal metal wire is carrying electric current from north to south direction. By applying uniform magnetic field, it is prevented from falling under gravity. The direction of this magnetic field should be towards

(1) West (2) North

(3) South (4) East

35. Two luminous point sources separated by a certain distance are at 10 km from observer. If the aperture of eye is 2.5×10^{-3} m and wavelength of light used is 600 nm, the distance of separation between point sources just seen to be resolved is

(1) 3.93 m (2) 1.33 m

(3) 2.93 m (4) 3.33 m

36. The de Broglie wavelength of electron in ground state of hydrogen atom, when radius of first orbit is 0.53 Å, is

(1) 1.33 Å (2) 3.33 Å

(3) 0.53 Å (4) 3.14 Å

37. A solid sphere of radius R has a charge Q distributed in volume with charge density $\rho = kr^a$, where k and a are constant and r is distance from centre. If the electric field magnitude at $r = \frac{R}{2}$ is $\frac{1}{8}$th times that at $r = R$. The value of a will be

(1) 9 (2) 8

(3) 6 (4) 2

38. A single slit diffraction pattern is formed using white light. For what wavelength of light, third secondary maximum in diffraction pattern coincides with the second secondary maxima in pattern for red light of wavelength 6500 Å?

(1) 4643 Å (2) 5216 Å

(3) 4848 Å (4) 3216 Å

39. Monochromatic light of wavelength 667 nm is produced by helium neon laser. The power emitted is 9 mW. The number of photons arriving per second on an average at a target irradiated by this beam is

(1) 9×10^{15} (2) 3×10^{12}

(3) 3×10^{16} (4) 3×10^{20}

40. Which of the following statements is incorrect?

(1) Pure silicon doped with trivalent impurity gives p-type semiconductor

(2) Majority charge carriers in p-type semiconductor are holes

(3) Minority charge carriers in n-type semiconductor are holes

(4) The resistance of intrinsic semiconductor increases with increase of temperature

41. A ray of light travelling in a transparent medium of refractive index μ falls on a surface separating medium from air at an angle of incidence of 45°. For which of the following value of μ, the ray can undergo total internal reflection?

(1) 1.1 (2) 1.25

(3) 1.33 (4) 1.50

42. An infinitely long metallic cylinder is kept parallel to uniform magnetic field B directed along positive z-axis. The direction of induced current on the surface of the cylinder as seen from z-axis will be
(1) Clockwise of positive z-axis
(2) Anticlockwise of positive z-axis
(3) Along magnetic field
(4) Zero

43. The apparent depth of a needle lying at the bottom of a tank, which is filled with water of refractive index 1.33 to a height of 12.5 cm is measured by microscope. If water is replaced by a transparent liquid of refractive index 1.63 up to same height, through what distance microscope have to be moved to focus on needle again?
(1) 2.9 cm
(2) 1.5 cm
(3) 1.73 cm
(4) 1.52 cm

44. An object is placed at a distance of 1.5 m from a screen and convex lens is between them. The magnification produced is 4. The focal length of the lens is
(1) 24 cm
(2) 36 cm
(3) 18 cm
(4) 28 cm

45. The focal length of the lenses of an astronomical telescope are 50 cm and 5 cm. The length of telescope when image is formed at the least distance of distinct vision is
(1) 54.16 cm
(2) 48.22 cm
(3) 55.0 cm
(4) 60.2 cm

CHEMISTRY

46. Oxidation state of sulphur in H_2SO_5 is
(1) +5
(2) +8
(3) +6
(4) +4

47. Which pair of solutions can form an acidic buffer?
(1) NaCl + HCl
(2) NaCl + CH_3COOH
(3) CH_3COOH + NaOH
(4) $CH_3COOH + H_2SO_4$

48. In which of the following equilibrium K_c and K_p are equal?
(1) $CO(g)+\frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$
(2) $2C(s)+O_2(g) \rightleftharpoons 2CO_2(g)$
(3) $2NO(g) \rightleftharpoons N_2(g)+O_2(g)$
(4) $CO(g)+3H_2(g) \rightleftharpoons CH_4(g)+H_2O(g)$

49. The solubility product of Ag_2CrO_4 is 1.1×10^{-12}. The solubility of Ag_2CrO_4 will be
(1) 6.5×10^{-5} mol/litre
(2) 1×10^{-6} mol/litre
(3) 6.5×10^{-4} mol/litre
(4) 1×10^{-3} mol/litre

50. The total kinetic energy of the molecules in 40 g Neon at T(K) is
(1) $\frac{3}{2}RT$
(2) 3 RT
(3) RT
(4) $\frac{RT}{2}$

51. Which of the following is antifertility drug?
(1) Bithionol
(2) Ofloxacin
(3) Norethindrone
(4) Equanil

52. Which is the monomer of natural rubber in the following?
(1) $CH_2 = \underset{\displaystyle Cl}{\overset{|}{C}} - CH = CH_2$
(2) $CH_2 = \underset{\displaystyle CH_3}{\overset{|}{C}} - CH = CH_2$
(3) $CH_2 = \underset{\displaystyle CH_3}{\overset{|}{C}} - CH_3$
(4) $CH_2 = CH - CH = CH_2$

53. The biodegradable polymer is
(1) Nylon-6
(2) PHBV
(3) PVC
(4) Buna-S

54. The essential amino acid among the following is
(1) Glutamic acid (2) Proline
(3) Tyrosine (4) Valine

55. Deficiency of vitamin C causes the disease
(1) Rickets (2) Scurvy
(3) Beri-Beri (4) Convulsions

56. Which has the maximum number of molecules among the following?
(1) 36 g $H_2O(l)$ (2) 64 g $CH_4(g)$
(3) 44 g $CO_2(g)$ (4) 10 g $H_2(g)$

57. The total energy associated with second energy shell of hydrogen atom is
(1) -2.18×10^{-18} J/atom
(2) -4.42×10^{-18} J/atom
(3) -0.545×10^{-18} J/atom
(4) -0.726×10^{-18} J/atom

58. The reaction for which $\Delta H > \Delta U$ at 298 K will be
(1) $H_2(g) + I_2(g) \rightarrow 2HI(g)$
(2) $PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$
(3) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
(4) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$

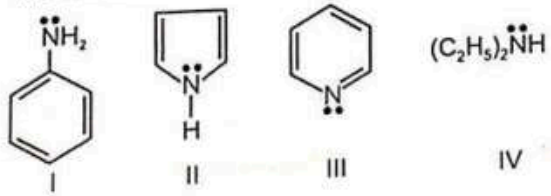
59. The gas produced from thermal decomposition of NH_4NO_3 is
(1) Nitrous oxide (2) Nitric oxide
(3) Ammonia (4) Nitrogen

60. The molecule which is not hydrolysed by water at 25°C is
(1) PCl_5 (2) CCl_4
(3) PCl_3 (4) BF_3

61. Chlorine on reaction with cold and dilute NaOH gives
(1) $NaClO_3 + NaCl$ (2) $NaOCl + NaClO_3$
(3) $NaCl + NaClO_4$ (4) $NaCl + NaOCl$

62. Which of the following ion is colourless?
(1) Sc^{3+} (2) Cr^{3+}
(3) Cu^{2+} (4) Mn^{2+}

63. The correct order of basicity of



I II III IV

in water is
(1) IV > III > I > II (2) IV > II > III > II
(3) IV > II > III > I (4) III > I > IV > II

64. In the following reaction

$$CH_3-CH_2-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow[NaOH]{Br_2}$$

The major product is
(1) $CH_3-CH_2-CH_2-NH_2$
(2) $CH_3-CH_2-NH_2$
(3) $CH_3-\overset{\overset{O}{\|}}{C}-H$
(4) CH_3-CH_2-Br

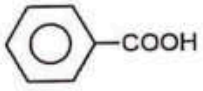
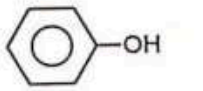
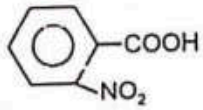
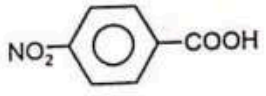
65. $A \xrightarrow[20\%\ H_2SO_4]{1\%\ HgSO_4} CH_3CHO$

The reactant 'A' is
(1) $CH_3C \equiv CH$ (2) $HC \equiv CH$
(3) $CH_2 = CH_2$ (4) CH_3-CH_3

66. $CH_3-CH_2-COOH \xrightarrow[(ii)\ H_2O]{(i)\ Br_2/Red\ phosphorus} (A)$

Compound (A) is
(1) $CH_3-\overset{\overset{O}{\|}}{C}-H$
(2) $CH_2(\overset{Br}{|})-CH_2-COOH$
(3) $CH_3-CH(\overset{Br}{|})-COOH$
(4) CH_3-CH_2-COBr

67. The order of acidity of compounds I – IV is

I. 
II. 
III. 
IV. 

(1) III > IV > I > II (2) IV > III > I > II
(3) III > I > IV > II (4) IV > I > III > II

68. EAN of cobalt in complex $[Co(CO)_4]$ is

(1) 36 (2) 35

(3) 37 (4) 34

69. Correct order of stability the given free radicals is

(1) $CH_3-\dot{C}H_2 > CH_3-\underset{CH_3}{\overset{CH_3}{\dot{C}}}-CH_3 > CH_3-\underset{CH_3}{\dot{C}H} > CH_2=CH-\dot{C}H_2$

(2) $CH_3-\underset{CH_3}{\overset{CH_3}{\dot{C}}}-CH_3 > CH_2=CH-\dot{C}H_2 > CH_3-\underset{CH_3}{\dot{C}H} > CH_3-\dot{C}H_2$

(3) $CH_2=CH-\dot{C}H_2 > CH_3-\underset{CH_3}{\overset{CH_3}{\dot{C}}}-CH_3 > CH_3-\underset{CH_3}{\dot{C}H} > CH_3-\dot{C}H_2$

(4) $CH_2=CH-\dot{C}H_2 > CH_3-\dot{C}H_2 > CH_3-\underset{CH_3}{\dot{C}H} > CH_3-\underset{CH_3}{\overset{CH_3}{\dot{C}}}-CH_3$

70. The IUPAC name for the following compound is

$CH_3-CH_2-\underset{\underset{CH_2}{\|}}{C}-CH_2-CH_2-CH_3$

(1) 3-propylbut-3-ene

(2) 4-ethylpent-4-ene

(3) 2-ethylpent-1-ene

(4) 2-propylbut-1-ene

71. Number of stereoisomers of complex $[Cr(NH_3)_3Cl_3]$ is

(1) Zero (2) 1

(3) 2 (4) 3

72. The weight of copper (At wt. = 63.5 g) displaced by a quantity of electricity which displaces 11200 mL of O_2 at STP will be

(1) 6.35 g (2) 63.5 g

(3) 31.75 g (4) 127 g

73. When initial concentration of the reactant is doubled, the half life period of a first order reaction

(1) Remains same

(2) Becomes halved

(3) Becomes doubled

(4) Becomes tripled

74. Cloud is a colloidal solution of

(1) Liquid in gas

(2) Gas in liquid

(3) Solid in gas

(4) Liquid in solid

75. The fraction of total volume occupied by the atoms present in a body centred cubic structure is

(1) $\frac{\pi}{3\sqrt{2}}$ (2) $\frac{\pi}{6}$

(3) $\frac{\sqrt{3}\pi}{8}$ (4) $\frac{\pi}{2\sqrt{2}}$

76. Which of the following is independent on temperature?

(1) Molarity

(2) Normality

(3) $\left(\frac{w}{v}\right)\%$

(4) Molality

77. A solution containing 18 g per dm^3 of glucose (molecular mass = 180 g mol^{-1}) is isotonic with 5% (w/v) solution of a non-volatile solute. The molecular mass of this non-volatile solute is

(1) 180 g mol^{-1} (2) 50 g mol^{-1}

(3) 500 g mol^{-1} (4) 250 g mol^{-1}

78. Which of the following pairs of compounds is isostructural?

(1) NH_3, BF_3 (2) I_3^-, XeF_2

(3) $BeCl_2$, ICl_3 (4) BCl_3, ClF_3

79. Decreasing order of stability of N_2, N_2^+ and N_2^- is

(1) $N_2^+ > N_2 > N_2^-$

(2) $N_2 > N_2^+ > N_2^-$

(3) $N_2 > N_2^- > N_2^+$

(4) $N_2^+ > N_2^- > N_2$

80. The correct shape and hybridisation for $XeOF_4$ are

(1) Pyramidal, sp^3d

(2) Octahedral, sp^3d^2

(3) Square pyramidal, sp^3d^2

(4) Trigonal bipyramidal, sp^3d

81. Which one of the following molecules does not contain π bond?

(1) SF_4 (2) SO_2

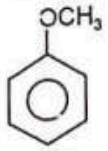
(3) SO_3 (4) CO_2

82. In which of the following species, bond angle is maximum?

(1) NO_2 (2) NO_2^+

(3) NO_2^- (4) NO_3^-

83. Which one of the following is most reactive towards electrophilic substitution reaction?

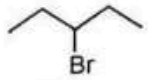
(1) 

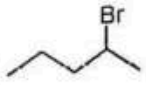
(2) 

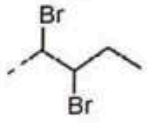
(3) 

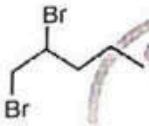
(4) 

84. The reaction of  with HBr produces

(1) 

(2) 

(3) 

(4) 

85. Benzene reacts with CH_3COCl in the presence of anhydrous $AlCl_3$ to form

(1) Benzyl chloride

(2) Toluene

(3) Acetophenone

(4) Benzophenone

86. Which one of following is not a common component of oxidising smog?

(1) Nitric oxide (2) O_3

(3) PAN (4) Freons

87. A mixture of CO and N_2 is called

(1) Synthesis gas (2) Water gas

(3) Producer gas (4) Laughing gas

88. On heating which of the following releases NO_2 most easily?

(1) $NaNO_3$ (2) $LiNO_3$

(3) KNO_3 (4) $RbNO_3$

89. Which one is most reactive towards S_N1 reaction?

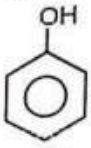
(1) C_6H_5Br (2) CH_3CH_2Br

(3) CH_3Br (4) $(CH_3)_3CBr$

90. 

$+ HI \longrightarrow A + B$

The products (A) and (B) are respectively

(1) 

and CH_3OH

(2) 

and CH_3I

(3) CH_3I and 

(4) 

and CH_3OH

BOTANY

91. Which of the following reaction is appropriate for nitrification performed by *Nitrococcus*?

(1) $2NO_2^- + O_2 \rightarrow 2NO_3^-$

(2) $2NH_3 + 3O_2 \rightarrow 2NO_2^- + 2H^+ + 2H_2O$

(3) $N \equiv N \rightarrow NH_3$

(4) $N_2 + O_2 \rightarrow 2NO$

92. Which of the following steps in sewage treatment removes the organic matter present in the effluent?

(1) Primary treatment

(2) Secondary treatment

(3) Tertiary treatment

(4) Sequential filtration

Akash Medical/IIT-JEE/Foundations

93. A 5-carbon ketose sugar is the primary CO_2 acceptor during carbon fixation in
(1) Maize and wheat
(2) *Chlorella* and *Spirogyra*
(3) Sorghum and rice
(4) Sugarcane and sorghum

94. Viroids differ from prions in having/being
(1) Low molecular weight RNA
(2) Infectious protein particle
(3) DNA molecule of low molecular weight without protein coat
(4) RNA molecule of high molecular weight with protein coat

95. How is our quality of life affected if, say, instead of 20,000 we have only 15,000 species of ants on earth? Answer to such naive questions have been explained to some extent through an analogy of rivet popper hypothesis by
(1) Paul Ehrlich (2) Humboldt
(3) Calvin (4) Schleiden

96. Foul odour, floral rewards as nectar and edible pollen grains, conspicuous look of flower etc. are required for
(1) Epihydrophily
(2) Anemophily
(3) Hypohydrophily
(4) Entomophily

97. Which of the following is the **correct** floral formula for the family, solanaceae?
(1) % δ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_{(3)}$
(2) % δ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_{1}$
(3) $\oplus\delta K_{(5)}\ \widehat{C_{(5)}A_5}\ \underline{G}_{(2)}$
(4) $\oplus\delta K_{(5)}\ C_{(5)}A_{(5)}\ \underline{G}_{(2)}$

98. Female and male gametophytes have an independent free living mostly photosynthetic existence in
(1) *Selaginella* (2) *Pinus*
(3) *Cycas* (4) *Ficus*

99. In each of the given set a conservation approach and an example of the method of conservation is given
(a) *In situ* conservation – National park
(b) *Ex situ* conservation – Wildlife sanctuary
(c) *In situ* conservation – Pollen bank
(d) *Ex situ* conservation – Botanical garden
Select the option with **correct** match of approach and method.
(1) (a) and (b)
(2) (a) and (c)
(3) (a) and (d)
(4) (b) and (d)

100. Which of the following is **incorrect** about archaebacteria?
(1) Methanogens are present in the gut of several ruminants.
(2) Methanogens produce biogas from the dung of cattle
(3) Some of the members are thermoacidophiles
(4) Cell wall structure is similar to eubacteria

101. Who first saw and described a live cell?
(1) Anton Von Leeuwenhoek
(2) Alexander von Humboldt
(3) Matthias Schleiden
(4) Robert Hooke

102. A mother is heterozygous for B-blood group and father is of AB-blood group. How many different phenotypes and genotypes respectively are possible for their sons?
(1) 4, 4 (2) 3, 2
(3) 3, 3 (4) 3, 4

103. Which of the following algae has fucoxanthin pigment and hydrocolloids in their cell wall?
(1) *Dictyota* (2) *Chara*
(3) *Polysiphonia* (4) *Porphyra*

104. Life cycle of *Eucalyptus* and *Cedrus* are
(1) Haplodiplontic
(2) Diplontic
(3) Haplontic
(4) Haplontic and diplontic respectively

105. Select the **incorrect** statement.
(1) Usually sclerenchyma are dead and without protoplasts
(2) Collenchyma may contain chloroplast
(3) Cells of collenchyma have thickening of ligno-suberin at the corners
(4) Phloem sclerenchyma (bast fibre) is absent in primary phloem

106. The T.S. of a plant specimen shows the following anatomical features under microscope.
(a) Endarch protoxylem
(b) Smaller vascular bundles are peripheral and large ones are centrally located.
(c) Pith absent
(d) Vascular bundles are conjoint and closed.
Identify the category of plant specimen.
(1) Dicot stem
(2) Dicot root
(3) Monocot root
(4) Monocot stem

107. "They do not have father but have a grandfather" this statement related to sex determination is **correct** for
(1) Chicks (2) Honey bee
(3) *Drosophila* (4) Butterflies

108. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Myotonic dystrophy	(i)	Down's syndrome
b.	Cystic fibrosis	(ii)	Autosomal dominant trait
c.	Colour blindness	(iii)	Autosomal recessive trait
d.	Characteristic palm crease	(iv)	X-linked trait

(1) a(i), b(iii), c(ii), d(iv)
(2) a(iv), b(iii), c(i), d(ii)
(3) a(ii), b(iii), c(iv), d(i)
(4) a(i), b(ii), c(iii), d(iv)

109. Besides the involvement of multiple genes for expression which of the following also takes into account the influence of environment?
(1) Starch grain size in pea
(2) Human skin colour
(3) Eye colour in *Drosophila*
(4) Pod colour in pea

110. Which option is appropriate for a group of related genera with still less number of similarities as compared to genus and species?
(1) Polymoniales
(2) Solanaceae
(3) Dicotyledonae
(4) *Triticum*

111. A lateral branch with short internodes and each node bearing a rosette of leaves and tuft of roots is found in
(1) Halophyte
(2) Xerophyte
(3) Hydrophyte
(4) Mesophyte

112. A large number of ovules are borne on central axis and septa are present, such placentation is found in
(1) *Dianthus* (2) Primrose
(3) China rose (4) *Argemone*

113. In the seeds of maize
(1) Seed coat is thick
(2) Endosperm is not bulky
(3) Embryo is large and situated in a groove
(4) Scutellum is shield shaped

114. Select the **correct** option for the given statements A and B.
A. Centrosome is an organelle without membrane and with triplet fibrils.
B. Centrioles in a centrosome lie parallel to each other like the cartwheel.
(1) Only A is correct
(2) Both A and B are correct
(3) Only B is correct
(4) Both A and B are incorrect

115. What will be the content of DNA in a somatic cell at metaphase if its meiotic products have 12 picogram DNA?

(1) 12 picogram (2) 24 picogram

(3) 36 picogram (4) 48 picogram

116. Select the **correct** match w.r.t. size of cellular component.

(1)	Perinuclear space	–	10-50 nm
(2)	Chloroplast	–	0.5-15 μ_m (length)
(3)	Mitochondria	–	Single membranous
(4)	Cisternae of Golgi complex	–	5-10 μ_m (diameter)

117. During anaphase

(1) Chromosomes are arranged at equatorial plate

(2) Each chromosome splits into four chromatids

(3) The centromeres are directed towards the equator

(4) Centromeres split and chromatids move to opposite poles.

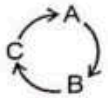
118. Three solutions are given

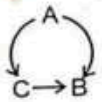
(i) ψ_w of solution A = –0.2 MPa

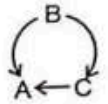
(ii) ψ_w of solution B = –0.1 MPa

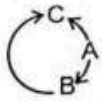
(iii) ψ_w of solution C= –0.15 MPa

What will be the direction of the movement of water when these solutions are separated by egg membrane?

(1) 

(2) 

(3) 

(4) 

119. The thick and elastic wall of the guard cell is seen towards the

(1) Subsidiary cell

(2) Stomatal aperture

(3) Outer side of it

(4) Outer as well as inner side of it

120. Out of more than 105 elements discovered so far, how many are essential for normal plant growth and development?

(1) 17 (2) 25

(3) 21 (4) 24

121. Cyclic photophosphorylation

(1) Occurs when light of wavelengths beyond 700 nm are available

(2) Occurs when only light of wavelengths 500 nm are available

(3) Occurs when only light of wavelengths below 680 nm are available for excitation

(4) Occurs when only light of wavelengths beyond 680 nm are available for excitation

122. What is the net gain of ATP when 4 molecules of acetyl CoA are completely oxidized during aerobic respiration?

(1) 60 (2) 72

(3) 48 (4) 30

123. Syngamy and triple fusion the events called double fertilization are unique to

(1) Angiosperms

(2) Only monocots

(3) Gymnosperms

(4) Lycopsida

124. Natural cytokinins are synthesized in regions like

(1) Mature leaf

(2) Developing shoot buds

(3) Ripened fruit

(4) Growing apices of the stems but not of roots

125. The nucleosomes in chromatin are seen as 'beads-on-string' structure when viewed under

(1) Simple microscope

(2) Compound microscope

(3) Electron microscope

(4) Magnifying glass after staining

126. DNA fingerprinting is a technique that

(1) Works on the principle of monomorphism of RNA

(2) Works on the principle of RNA polymorphism

(3) Finds the variations in individuals at protein level

(4) Has immense application in the field of forensic science

127. A piece of DNA if needed to be propagated during recombinant DNA procedures, requires a vector which provides the
(1) Energy because replication is very expensive process
(2) Origin of replication
(3) Transcription unit
(4) Selectable marker which eliminates the transformants

128. The valine is coded by GUU GUC GUA GUG whereas glycine is coded by GGU GGA GGC GGG, it reflects the genetic code is
(1) Degenerate (2) Universal
(3) Not universal (4) Non-ambiguous

129. Which of the following variety is **correctly** matched w.r.t. its crop?

(1)	Pusa Swarnim	–	Cauliflower
(2)	Pusa Komal	–	Wheat
(3)	Pusa Gaurav	–	Rapeseed mustard
(4)	Pusa Sawani	–	Flat bean

130. Vertical distribution of different species of herbs, shrubs and trees occupying different levels in tropical rain forests is called
(1) Fragmentation
(2) Ecological succession
(3) Stratification
(4) Scarification

131. Which of the following statements is **incorrect** w.r.t. biological magnification of DDT in an aquatic food chain?
(1) DDT is increased at successive trophic levels
(2) Disturbs magnesium metabolism causing thinning of egg cell
(3) Disturbs calcium metabolism causing thinning of egg shell
(4) Causing decline in bird population

132. Match the columns and select the **correct** option.

	Column-I		Column-II
a.	Catalytic converters	(i)	Particulate matter
b.	Electrostatic precipitator	(ii)	Reduce emission of poisonous gases
c.	Green muffler scheme	(iii)	Reduce noise pollution
d.	Sanitary landfills	(iv)	Solid waste disposal

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)
(3) a(ii), b(i), c(iii), d(iv) (4) a(i), b(ii), c(iv), d(iii)

133. Which of the following are attributes of population?
(1) Species richness and species interaction
(2) Births and deaths
(3) Species evenness and sex
(4) Sex ratio and birth rates

134. The bacteria that play a great role in recycling nutrients like nitrogen, iron, sulphur etc; belong to which mode of nutrition?
(1) Photoautotrophic
(2) Chemoautotrophic
(3) Heterotrophic
(4) Holozoic

135. Fruit production without fertilization is known as
(1) Apomixis
(2) Parthenogenesis
(3) Parthenocarpy
(4) Pseudocarpic

ZOOLOGY

136. Choose the **correct** statement w.r.t. human axial skeleton.
(1) Thoracic vertebrae and sternum both are present in ventral side of rib cage
(2) Spinal cord passes through neural canal which are present in centra of vertebrae
(3) Human skull articulates with first cervical vertebra called axis
(4) Bones of cranium included 2 paired bones (parietal and sphenoid) and 6 unpaired bones (ethmoid, frontal, nasal, temporal, occipital, zygomatic)

137. How many ribs articulate directly with vertebral column on their dorsal side?
(1) 12 (2) 24
(3) 7 (4) 14

138. Which one is the best distinguishing feature of biceps?
(1) Involuntary muscle fibres with structural syncytium
(2) Striated appearance, voluntary control
(3) Cells are branched, cylindrical with multi-nucleated condition
(4) Fusiform shape with smooth appearance

139. In which of the following option genus name, its two characters and its phylum are **correctly** matched?

	Genus name		Two characters	Phylum
(1)	*Sepia*	a	Presence of water vascular system	Mollusca
		b	External fertilisation	
(2)	*Saccoglossus*	a	Worm like, marine, segmented worm	Hemichordata
		b	Possess stomochord	
(3)	*Salpa*	a	Presence of ventral heart	Chordata
		b	Notochord in only larval tail	
(4)	*Antedon*	a	Chitinous shell covers the body	Echinodermata
		b	Spiny skinned	

140. Choose the **correct** match w.r.t. *Periplaneta*

(1)	Gizzard	–	Storage of food
(2)	Gastric caeca	–	Present at the junction of midgut and hindgut
(3)	Heart	–	13 chambered, present on ventral side of body
(4)	Anal cerci	–	Paired outgrowth from 10th abdominal segment

141. How many structures from given below in the box are present in thorax region of male cockroach?

Anal cerci, Anal style, Tegmina, Anterior aorta, Vestibulum

(1) One
(2) Two
(3) Three
(4) Four

142. Read the following statements and choose the option with only **correct** statements.
(a) *Exocoetus* lack the capacity to regulate their body temperature.
(b) *Hyla* possess cloaca which opens to outside the body.
(c) *Bungarus* possess external ear openings
(d) *Myxine* is devoid of scales and paired fins
(1) (a), (b) and (c) (2) (b), (c) and (d)
(3) (a), (b) and (d) (4) (a), (c) and (d)

143. Air bladder provides buoyancy in
(1) *Clarias* (2) *Petromyzon*
(3) *Ichthyophis* (4) *Pristis*

144. Corpora quadrigemina is present in
(1) Lateral side of forebrain
(2) Dorsal side of midbrain
(3) Ventral side of hindbrain
(4) Roof of diocoel

145. A structure, lies at the base of the thalamus, contains centre for all **except**,
(1) Gastric secretions
(2) Body temperature
(3) Urge of eating
(4) Urge of drinking

146. Choose the **incorrect** statement w.r.t. structure of human eye.
(1) The fovea is a thinned-out portion of retina where only cones are densely packed
(2) The diameter of pupil is regulated by the muscle fibres of Iris
(3) Aqueous humour is present between lens and the retina
(4) The rods contain photopigments rhodopsin or visual purple

147. At the base of the cochlea, scala vestibuli ends at the A, while the scala tympani terminates at the B

Select the **correct** option for A and B.

	A	B
(1)	Round window	Oval window
(2)	Oval window	Round window
(3)	Ear drum	Round window
(4)	Ear drum	Oval window

148. Select the **correct** match w.r.t. source gland with its respective hormone as well as the function.

	Source gland	Hormone	Function
(1)	Neurohypophysis	ADH	Reabsorption of water from latter part of renal tubules
(2)	Thyroid gland	TCT	Regulates the blood Ca^{+2} levels
(3)	Pars distalis	MSH	Stimulation of melanocytes
(4)	Acini of pancreas	Glucagon	Carbohydrate metabolism

149. A gland 'P' secretes two hormone 'R' and 'S' which regulate the basal metabolism in our body. Over-secretion of 'R' and 'S' from 'P' causes a disease which is characterised by protrusion of the eyeballs and enlargement of gland 'P'. This disease is possibly

(1) Addison's disease (2) Graves' disease
(3) Simple Goitre (4) Acromegaly

150. The hormones of pancreas

(1) Are hydrophobic in nature
(2) Interact with membrane – bound receptors and generate secondary messengers for their action
(3) Enter inside the cell and form hormone receptor complex inside the nucleus
(4) Are iodothyronines and hydrophilic in nature

151. Urea is the chief nitrogenous excretory product of

(1) Aquatic amphibians (2) Marine fishes
(3) Aquatic insects (4) Bony fishes

152. Read the following statements and choose the **correct** answer.

A : Each kidney of an adult human measures 5-7 cm in length and 10-12 cm in width.

B : The Malpighian corpuscles, PCT and DCT of the nephron are situated in the cortical region of the kidney whereas the loop of Henle dips into the medulla.

(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

153. Which of the following parts of nephron are helpful in the maintenance of ionic and acid base balance of the body fluids by secretion of H^+, K^+ and ammonia?

(1) PCT, Loop of Henle
(2) PCT, DCT, Collecting duct
(3) DCT, Loop of Henle
(4) DCT, Bowman's capsule

154. Given below are four statements (a-d) regarding human circulatory system.

(a) Human heart is protected by a double walled membranous bag, pericardium.
(b) Person with blood group B can donate the blood to any person with AB or O blood group under ABO blood group system
(c) Chordae tendineae are present in only ventricles
(d) Hypertension is the sustained elevation of blood pressure from its normal range

From above given statements choose the option with only **incorrect** statement(s).

(1) (a) and (b) (2) (b) and (c)
(3) Only (b) (4) (b) and (d)

155. Calculate the value of stroke volume for a person whose cardiac output is about 5040 ml/min and heart rate is 72 / minute

(1) 70 ml
(2) 80 ml
(3) 90 ml
(4) 100 ml

156. During normal inspiration,
(1) Diaphragm and the inter-costal muscles become relax
(2) Volume of thoracic cavity increases
(3) Intra-pulmonary pressure slightly exceed the atmospheric pressure
(4) 5000 ml air from outside moves into the lungs in each breadth

157. Choose the **mismatch** w.r.t. respiratory volumes/capacities
(1) FRC = ERV + EC
(2) TLC = VC + RV
(3) EC = VC – IRV
(4) ERV = FRC – RV

158. Nucleases act on nucleic acids in
(1) Pancreas
(2) Small intestine
(3) Buccal cavity
(4) Stomach

159. Which of the following cells of gastric mucosa play an important role in erythropoiesis?
(1) Goblet cells
(2) Chief cells
(3) Oxyntic cells
(4) Paneth cells

160. Match the column I with column II and choose the **correct** answer.

	Column-I		Column-II
a.	IgA	(i)	Mediator in allergic responses
b.	IgD	(ii)	Have a role in activating and suppressing lymphocyte activity
c.	IgE	(iii)	Can cross the placenta and confer immunity to the foetus
d.	IgG	(iv)	Present in colostrum

(1) a(iv), b(ii), c(iii), d(i)
(2) a(iv), b(ii), c(i), d(iii)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(iv), b(i), c(iii), d(ii)

161. Select the **correct** statement.
(1) Barbiturates are very effective pain killer and are very useful for patients who undergo surgery
(2) Nicotine stimulates adrenal cortex to release adrenaline and nor-adrenaline into blood circulation
(3) Use of Cocaine, makes a person more wakeful, alert, active and cause excitement
(4) Cannabinoids interferes with the transport of the neuro-transmitter dopamine

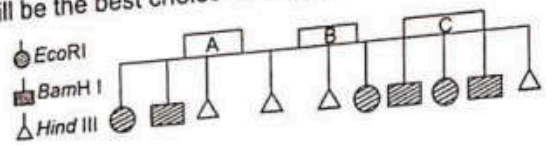
162. Which one of the following option gives the **correct** matching of a disease with its causative organism and mode of infection?

	Disease	Causative organism	Mode of infection
(1)	Ascariasis	*Ascaris*	Bite of infected female *Culex* mosquito
(2)	Typhoid	*Salmonella typhi*	Droplet infection
(3)	Amoebic dysentery	*Entamoeba histolytica*	Contaminated food and water
(4)	Malaria	*P. vivax*	Bite of larva of *Anopheles* mosquito

163. Which of the following would be an advantage of interspecific hybridisation?
(1) It is useful in overcoming inbreeding depression
(2) It helps in evolve a pure line
(3) It increases homozygosity
(4) Progeny contains desirable features of both parents which have considerable economic value

164. Silver revolution is associated with enhancement of
(1) Egg production
(2) Milk production
(3) Fish production
(4) Wheat and rice production

165. If only gene C without any flanking sequence is to be isolated from given fragment of DNA, what will be the best choice of enzyme?



(1) *Eco*RI
(2) *Bam*H I
(3) *Hind* III
(4) Both *Eco*RI and *Bam*H I

166. Read the following statements w.r.t. PCR.

(a) *Taq* polymerase added a nucleotide to the 3'-OH end of a growing strand.
(b) Primers bind to their complementary sequences on the single-stranded templates between temperature 70°C to 80°C in annealing step.
(c) 1 billion copies of a DNA fragment are made at the end of 30 PCR cycles.

From above
(1) (a) and (b) are correct, (c) is incorrect
(2) (a) and (c) are correct, (b) is incorrect
(3) (b) and (c) are correct, (a) is incorrect
(4) (a), (b) and (c) all are correct

167. Insertional inactivation in selectable marker which produce colour in chromogenic substance, will
(1) Produce blue colour colony of recombinants
(2) Produce white colour colony of non-recombinants
(3) Produce white coloured colony of recombinants
(4) Produce colourful colony of both recombinants and non-recombinants

168. Why adenosine deaminase **can't** be prescribed orally to a SCID patient?
(1) It may increase the pH of blood
(2) It can be digested in small intestine
(3) It will absorb in small intestine
(4) It is bitter in taste so very difficult to swallow

169. Select the **mismatch.**

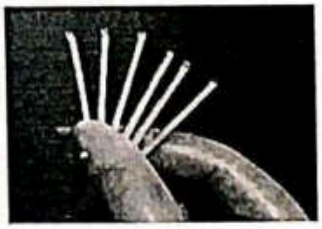
(1)	ELISA	–	Serve the purpose of early diagnosis
(2)	Gene therapy	–	Allow correction of a gene defect
(3)	Golden rice	–	Vitamin A enriched rice
(4)	*Agrobacterium*	–	Useful as a vector for both monocot and dicot plants

170. Placental mammal, anteater, and Australian marsupial numbat, have evolved to share many similar characteristics. This type of evolution be referred to as
(1) Adaptive divergency
(2) Divergent evolution
(3) Convergent evolution
(4) Adaptive radiation

171. A gene locus has two alleles 'A' and 'a'. If the frequency of dominant allele A is 0.6 then what will be the frequency of heterozygous?
(1) 0.36 (2) 0.16
(3) 0.48 (4) 0.98

172. Homologous structures are a result of
(1) Divergent evolution
(2) Adaptive convergence
(3) Co-evolution
(4) Convergent evolution

173. Identify the contraceptive device depicted in the figure given below and select the **incorrect** statement regarding it



(1) It contains either progestogens alone or combination of progestogens with estrogens
(2) Implanted under the skin for contraception
(3) It inhibits only ovulation compare to pill which inhibit both ovulation and implantation
(4) It is effective for longer (3 to 5 years) time period

174. Semen collected either from husband or a healthy donor is artificially introduced into vagina in
(1) ET
(2) AI
(3) IUT
(4) ZIFT

175. Number of chromosomes in meiocyte of cat is
(1) 78
(2) 38
(3) 12
(4) 46

176. Select the **odd** one w.r.t. hermaphrodite animals
(1) Earthworm
(2) *Periplaneta*
(3) Leech
(4) Sponges

177. The process spermiogenesis is
(1) Formation of primary spermatocytes from spermatogonium
(2) Transformation of spermatid into sperm
(3) Release of sperm head from seminiferous tubules
(4) Formation of spermatid from secondary spermatocytes

178. Which of the following cell undergoes meiosis II during gametogenesis?
(1) Oogonium
(2) Primary spermatocyte
(3) Secondary spermatocyte
(4) Spermatogonium

179. Cuboidal epithelium is present in
(1) Inner lining of fallopian tube
(2) Inner lining of brain ventricles
(3) Thyroid follicles
(4) Trachea and bronchi

180. Read the following statements and choose the **correct** answer.
A : In a primary structure of protein, the first amino acid is called as N-terminal amino acid while the last amino acid is called the C-terminal amino acid.
B : In a polysaccharide like glycogen, the right end is called reducing end while left end is called non-reducing end.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

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